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Centre Number	Index Number	Name

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CHEMISTRY **5070/02**

Paper 2 Theory

October/November 2005

1 hour 30 minutes

Candidates answer on the Question Paper.
Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in blue or black pen.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.
Write your answers in the spaces provided on the Question Paper.

Section B
Answer any **three** questions.
Write your answers on any lined pages and/or separate answer paper.

You may use a calculator.
You may use a pencil for any diagrams, graphs, or rough working.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.
A copy of the Periodic Table is printed on page 16.

For Examiner's Use	
Section A	
B8	
B9	
B10	
B11	
Total	

This document consists of **13** printed pages and **3** lined pages.

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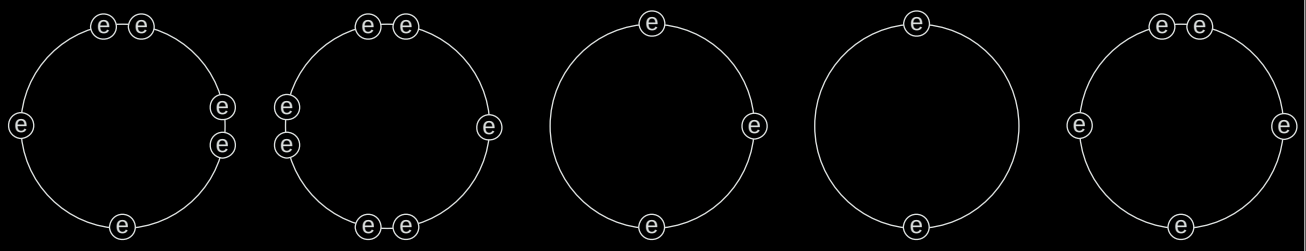
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Section A

Answer **all** the questions in this section in the spaces provided.

1 These diagrams show the electron arrangement in the outer shells of five elements, **A** to **E**. All elements are from Period 3 of the Periodic Table.



A **B** **C** **D** **E**

(a) Put the letters **A** to **E** in the table to show which elements are metals and which are non-metals.

	metals	non-metals
elements		

[2]

(b) Which element is most likely to be in Group VI?

..... [1]

(c) Which element will form an ion of the type X^{2+} ?

..... [1]

(d) Which element has an atomic number of 15?

..... [1]

(e) Which two elements will form an ionic compound with a formula of the type YZ_2?

..... [1]

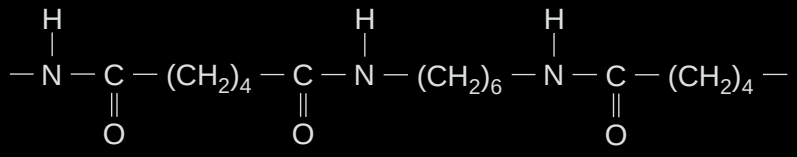
[Total: 6 marks]

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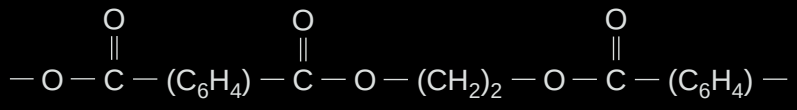
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2 These diagrams show sections of the polymer chain of two condensation polymers.

Nylon



Terylene



(a) **(i)** Draw a circle around an amide linkage in the diagram. Label this **amide**.

(ii) Draw a circle around an ester linkage in the diagram. Label this **ester**. [2]

(b) Name a type of naturally occurring polymer that has a similar linkage to nylon.

..... [1]

(c) The formulae of the two monomers used to make nylon are shown below.

Nylon monomers $HOOC(CH_2)_4COOH$ $H_2N(CH_2)_6NH_2$

Deduce the formulae of the two monomers that are used to make *Terylene*.

***Terylene* monomers**

[2]

(d) Sea fishing nets used to be made from natural fibres. Many nets are now made from nylon. Suggest one **advantage**, other than strength, and one **disadvantage** of using nylon rather than natural fibres to make sea fishing nets.

advantage .....

disadvantage ..... [2]

[Total: 7 marks]

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3 This table shows the soil pH ranges required by different crops for growth.

crop	pH range
peanut	5.0 – 6.5
millet	6.0 – 6.5
sunflower	6.0 – 7.5
paprika	7.0 – 8.5
mango	5.5 – 6.0

(a) A farmer plants peanut and millet crops. Only the peanut crop grows well. Predict the pH of the soil.

..... [1]

(b) Which other crop is most likely to grow well in the same soil?

..... [1]

(c) The farmer adds calcium hydroxide, $Ca(OH)_2$, and ammonium sulphate, $(NH_4)_2SO_4$, to the soil.

Explain the purpose of using each compound.

.....

.....

..... [3]

(d) A reaction occurs between calcium hydroxide and ammonium sulphate.

(i) Complete the equation for this reaction.

$Ca(OH)_2 + (NH_4)_2SO_4 \rightarrow$ …… + …… + $2H_2O$

(ii) Explain why the farmer should not have added these two compounds to the soil at the same time.

..... [3]

[Total: 8 marks]

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4 This table shows some information about two homologous series; the alkanes and t chlorides.

alkanes	**acid chlorides**	
formula	name	formula
C_2H_6	ethanoyl chloride	CH_3COCl
C_3H_8		C_2H_5COCl
C_4H_{10}	butanoyl chloride	C_3H_7COCl
C_5H_{12}	pentanoyl chloride	C_4H_9COCl

Use the information in the table to answer the following questions.

(a) Name the acid chloride with the highest boiling point.

..... [1]

(b) Deduce the name of the acid chloride with the formula C_2H_5COCl.

..... [1]

(c) The general formula for alkanes is $C_xH_{(2x + 2)}$.
Deduce the general formula for acid chlorides.

..... [1]

(d) **(i)** Name the products of the complete combustion of an alkane.

.....

(ii) Would you expect the products of complete combustion of the acid chlorides to be the same as in **(i)**? Explain your reasoning.

.....
[2]

[Total: 5 marks]

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5 An experiment was carried out to measure the rate of reaction between excess po calcium carbonate and dilute acids.

(a) In **Experiment 1**, 25 cm^3 of 1.5 mol/dm^3 hydrochloric acid was used.

Complete the equation for the reaction by filling in the missing state symbols.

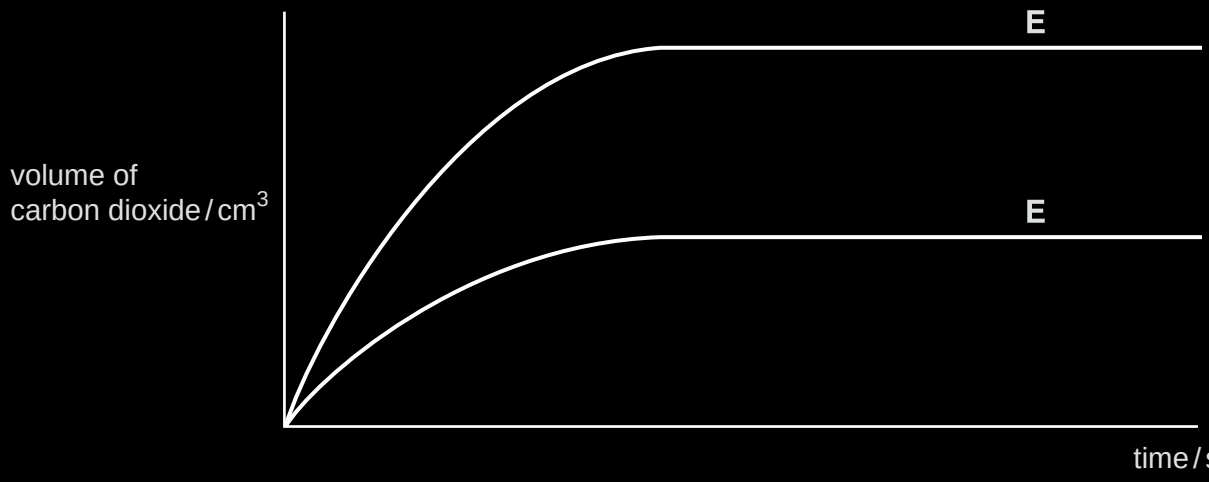
(i) $2HCl(\ldots\ldots\ldots) + CaCO_3(\ldots\ldots\ldots) \rightarrow CaCl_2(aq) + H_2O\ (\ldots\ldots\ldots) + CO_2(\ldots\ldots\ldots)$

(ii) Calculate the total volume of carbon dioxide that is made from this reaction at r.t.p.

[4]

(b) A further experiment using hydrochloric acid, **Experiment 2**, was carried out.

The results of **Experiments 1** and **2** are shown on the graph.



Suggest the **concentration** and **volume** of acid used for **Experiment 2**.

concentration …… mol/dm^3

volume …… cm^3 [2]

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(c) **Experiment 3** was carried out using 25 cm^3 of 1.5 mol/dm^3 sulphuric acid.
The initial rate of reaction for **Experiment 3** was faster than for the other experim
but the reaction stopped suddenly after only a small amount of gas had been given of

(i) Name the salt formed in **Experiment 3.**

.....

(ii) Explain why the reaction stops suddenly.

.....

(iii) Explain why the initial rate of reaction was faster than for the other experiments.

.....
[4]

[Total: 10 marks]

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6 The table below shows some information about two copper ores, tenorite and cuprite. Both contain copper oxide.

ore	formula of copper oxide in ore	oxidation number of copper	percentage of copper by mass
tenorite		+2	80.0%
cuprite	Cu_2O		

(a) **(i)** What is the formula of the copper compound in tenorite?

.....

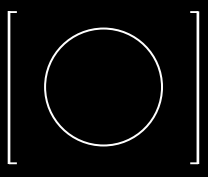
(ii) What is the oxidation number of copper in cuprite, Cu_2O?

.....

(iii) Calculate the percentage of copper by mass in Cu_2O.

[5]

(b) Another ore of copper contains copper(II) sulphide.
Complete the dot and cross diagram below for copper(II) sulphide showing outer electrons only.



[2]

[Total: 7 marks]

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7 An oil refinery uses two different processes, **Process 1** and **Process 2**, to crack n
The table below shows some information about the percentage yields of products from
process.

product	yield / %	
	Process 1	**Process 2**
hydrogen	1	1
methane	18	14
ethene	32	20
propene	13	15
C_4 hydrocarbons	9	10
C_5 to C_8 hydrocarbons	27	40

The refinery sells ethene and C_5 to C_8 hydrocarbons.
Ethene is used to make addition polymers, and C_5 to C_8 hydrocarbons are added to petrol.
Use the information given to explain why the refinery must use **both** processes to meet the high demand for both ethene and C_5 to C_8 hydrocarbons.

.....

..... [2]

[Total: 2 marks]

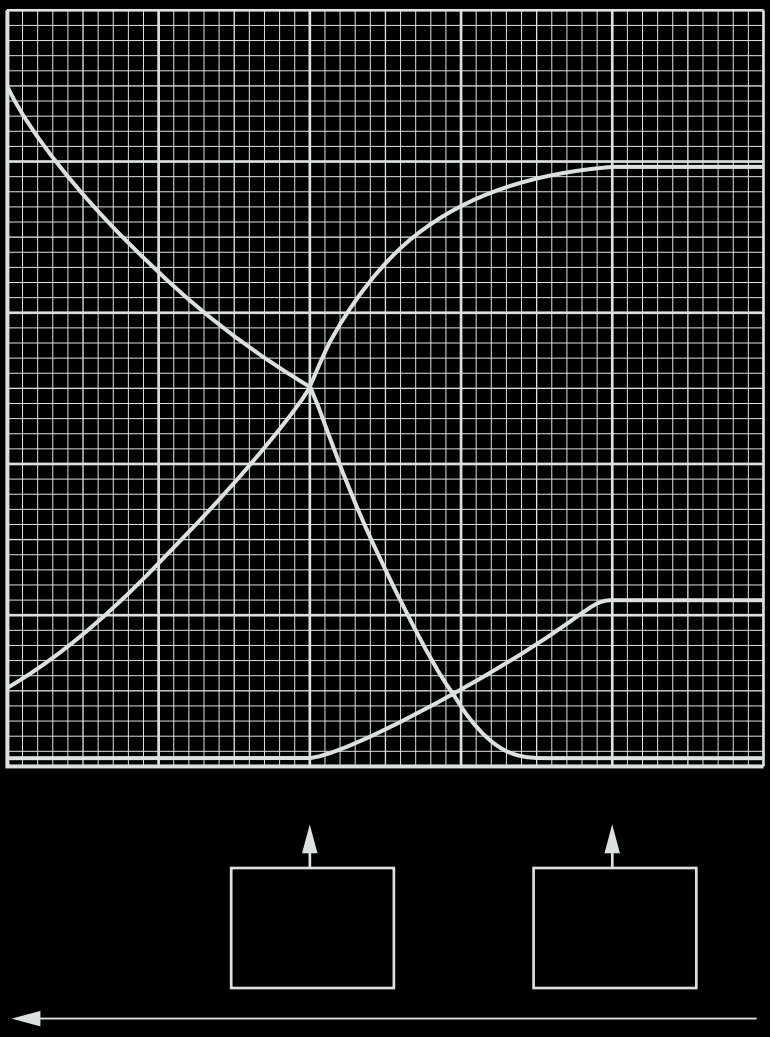
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Section B

Answer **three** questions from this section.

Write your answers on any lined pages provided and/or a separate answer paper.

B8 This graph shows how the percentage of three of the gases in the Earth's atmosphere has changed over five thousand million years.



Use information from the graph to answer the following questions.

(a) **(i)** How long have the percentages of all gases in the atmosphere remained unchanged?

(ii) Name **gas 3**. Give a reason for your answer. [3]

(b) **(i)** Describe how the percentages of carbon dioxide and oxygen have changed.

(ii) Suggest an explanation for the changes that have taken place in carbon dioxide and oxygen percentages, identifying the processes involved and giving equations for any reactions. [5]

(c) Oxygen is separated from air by fractional distillation. Outline how this separation takes place. [2]

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B9 Chlorine, bromine and iodine are elements in Group VII of the Periodic Table.

(a) Describe how you would carry out a series of experiments to show the trend in reac these three elements, using the reagents shown below.

aqueous chlorine	aqueous potassium chloride
aqueous bromine	aqueous potassium bromide
aqueous iodine	aqueous potassium iodide

Your answer should include details of

- which of the reagents you would use in each experiment,
- a table showing the observations you would expect to see,
- the equations for any reactions. [7]

(b) Chlorine reacts with water to make a solution that can be used as a bleach.
The equation is shown below.

$$Cl_2 + H_2O \rightarrow HCl + ClOH$$

Use oxidation numbers to show that chlorine is both oxidised and reduced in this reaction. [3]

[Total:10 marks]

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B10 A student carried out an electrolysis of dilute sulphuric acid and collected the gases

(a) Draw a labelled diagram to show the apparatus used.

(b) **(i)** Give the formulae of all the ions present in the solution.

(ii) Write half equations for the reactions at the anode and cathode. Use the half equations to construct an overall equation for the reaction and give tests for any gases evolved.

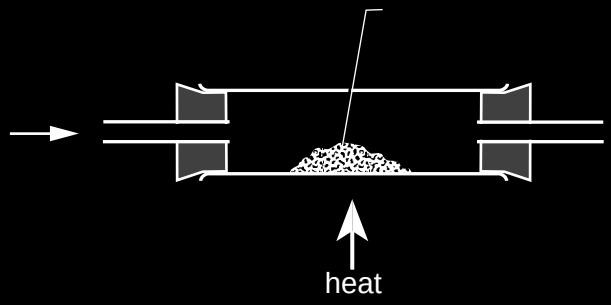
(iii) Use your equations to explain how the composition of the solution changes after the electrolysis has been running for some time.

[6]

(c) Describe another method for making hydrogen from dilute sulphuric acid. Your answer should include names of the reagents you use and an equation for the reaction. [2]

[Total: 10 marks]

B11 The diagram below shows an experiment in which steam was passed over hot iron filings. The products of the reaction are iron oxide, Fe_3O_4, and a gas which burns with a blue flame.



(a) Write an equation, including state symbols, for the reaction and describe what you would see as the iron reacts with the steam. [3]

(b) Describe how the observations would be different if the experiment was repeated using each of the following two metals in place of the iron filings.

(i) magnesium

(ii) copper

[3]

(c) **(i)** Both copper and aluminium are good conductors of electricity. Explain why overhead cables are usually made from aluminium and not copper.

(ii) Draw a diagram to show the structure and bonding of aluminium metal. Use your diagram to explain why aluminium conducts electricity so well.

[4]

[Total: 10 marks]

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DATA SHEET
The Periodic Table of the Elements

Group I	Group II										Group III	Group IV	Group V	Group VI	Group VII	Group 0
							1 H Hydrogen 1									4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4										11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12										27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulphur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89 †															

58-71 Lanthanoid series
90-103 Actinoid series

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60
232 Th Thorium 90	Pa Protactinium 91	238 U Uranium 92

Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63
Np Neptunium 93	Pu Plutonium 94	Am Americium 95

157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66
Cm Curium 96	Bk Berkelium 97	Cf Californium 98

165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69
Es Einsteinium 99	Fm Fermium 100	Md Mendelevium 101

173 Yb Ytterbium 70	175 Lu Lutetium 71
No Nobelium 102	Lr Lawrencium 103

Key

a	a = relative atomic mass
X	X = atomic symbol
b	b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

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